10

Linking growth models of different resolutions

10.1 Introduction

Growth models have both apparent and functional resolution (Ritchie and Hann, 1997a). *Apparent* resolution is the maximum required level of resolution as determined by the model's required inputs. *Functional* resolution is determined by how the model operates and behaves. Both the apparent and functional resolution of a growth and yield model have significant implications for its accuracy. For example, individual-tree growth and yield models can provide unbiased estimates of growth, but can be less accurate in projecting mortality when compared to a stand-level approach. This suggests that there are important advantages and disadvantages of each approach to growth modeling that need to be recognized by both modelers and model users.

Most models have the same apparent and functional resolutions, but there are significant efforts to link growth models of different resolutions in an attempt to avoid the shortcomings of any single approach. The advantages of a linked model are that it can significantly reduce projection errors, constrain predictions, and maintain multiple resolutions. Although there are several different examples of linked growth models, this chapter will focus on the two most common classes: (1) linked stand-level and size-class models and (2) linked stand- and individual-tree-level models.

10.2 Linked stand- and size-class models

Stand-level growth models project changes in total basal area, number of trees, and volume, while size-class models maintain frequencies of individuals by diameter at breast

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height (DBH) classes in their predictions. Stand-level models are relatively simple and have low dimensionality, but can provide highly accurate predictions of both growth and mortality. Size-class models have the ability to represent variability of individuals in a stand, which is important for understanding response to management and stand structure. Two primary methods are used for linking stand and size-class models: (1) the use of parameter recovery techniques; and (2) modified stand table projections.

10.2.1 Parameter recovery

Parameter recovery works by providing a parameter estimator of a probability or cumulative distribution that is a function of stand-level variables (Figures 10.1 and 10.2).

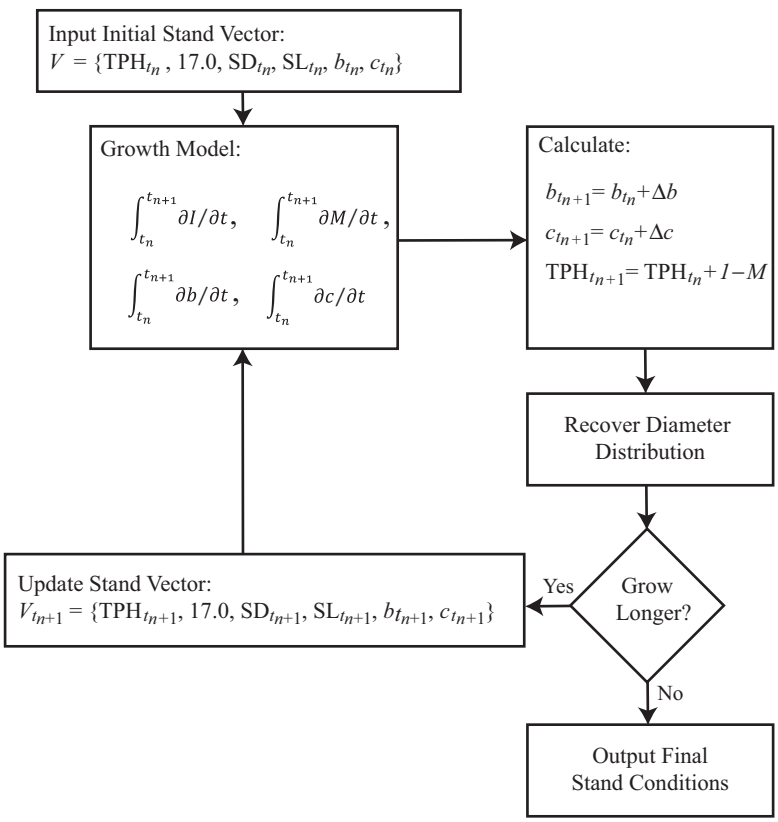


Figure 10.1 Flowchart of parameter prediction model proposed by Hyink and Moser (1983). V is total volume ($m^3\ ha^{-1}$), TPH is stem density (no. ha^{-1}), SD is sum of the tree diameters, SL is sum of the tree diameters larger than 18 cm, b is the Weibull scale parameter, c is the Weibull shape parameter, I is ingrowth (no. ha^{-1}), and M is mortality (no. ha^{-1}).

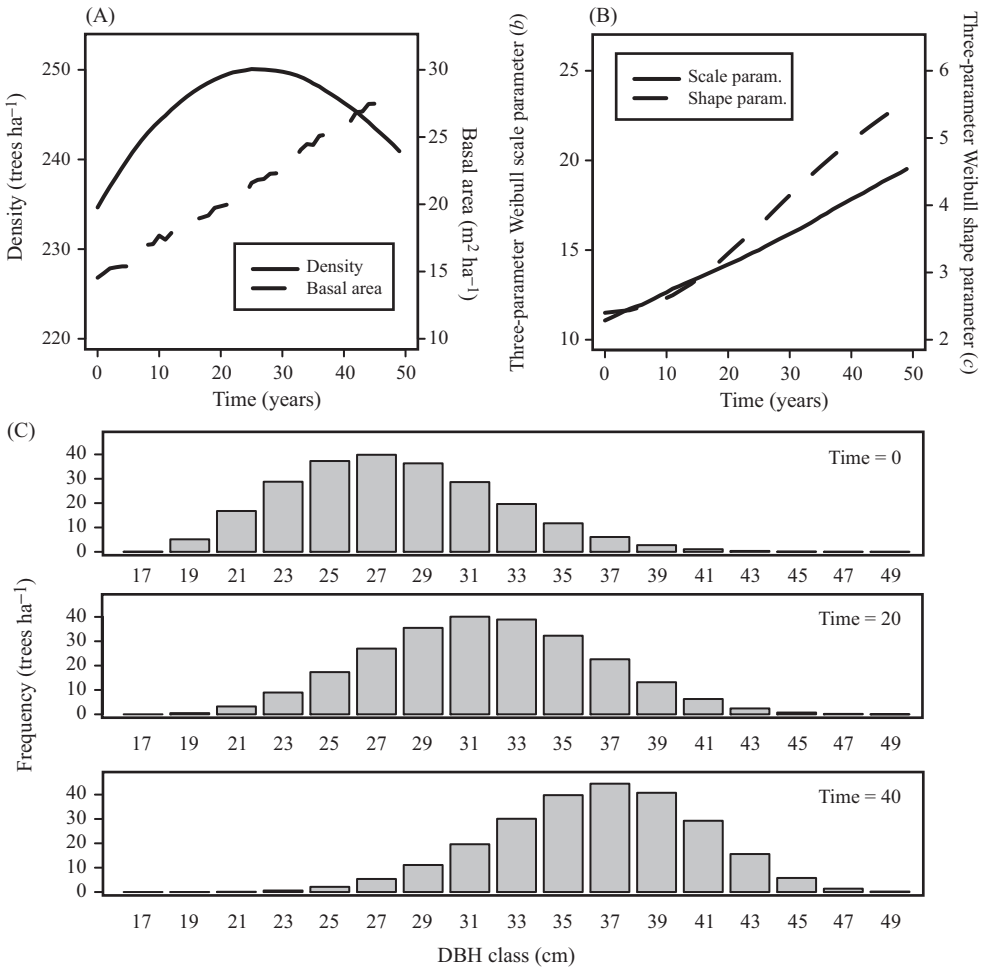


Figure 10.2 Simulation results from the parameter prediction model proposed by Hyink and Moser (1983): (A) Number of trees (ha^{-1}) and basal area (m^2 ha^{-1}) over time; (B) the Weibull scale and shape parameters over time; (C) recovered diameter distributions for three points in time based on projected density, scale, and shape parameters.

The two distributions most commonly used with parameter recovery estimators are the Weibull (Hyink and Moser, 1983) and Johnson's Sb (Knoebel and Burkhart, 1991; Fonseca *et al.*, 2009) distributions. These estimators can be both moment- (Burk and Newberry, 1984) and percentile- (Bailey *et al.*, 1989) based. In the moment-based approach, the parameters are estimated from the first-, second- and third-order moments of the diameter distribution (i.e. mean, variance, and skewness). In the percentile-based approach, the parameters are recovered from various percentiles of the diameter distribution.

For example, Bailey *et al.* (1989) used the quadratic mean diameter, minimum diameter, and the 25th, 50th, and 95th percentiles to estimate parameters of the Weibull distribution.

A hybrid approach that uses both moment- and percentile-based estimators was successfully applied by Baldwin and Feduccia (1987). Although the percentile-based and hybrid approaches perform better than the moment-based approach (e.g. Cao, 2004), moments are the preferred method in growth and yield models because they ensure numeric compatibility and generally require fewer equations. In addition, percentile methods have several other disadvantages such as not providing an unique solution, difficulty in selecting appropriate percentiles, and limited biological interpretability (Fonseca *et al.*, 2009).

Since most stand-level growth models do not predict variance or skewness of diameter distributions through time, alternative estimators are used. For example, Cao *et al.* (1982) iteratively estimated the Weibull parameters based on minimum, arithmetic mean, and quadratic mean diameters:

$$\begin{aligned} a &= 0.5 \cdot \mathrm{DBH}_{\mathrm{Min}} \\ b &= \frac{(\overline{\mathrm{DBH}} - a)}{\Gamma\left(1 + \frac{1}{c}\right)} \\ 0 &= \mathrm{QMD}^2 + a^2 - 2 \cdot \overline{\mathrm{DBH}} \cdot \left(\Gamma\left(1 + \frac{2}{c}\right)\right) \end{aligned} \qquad [10.1]$$

where a, b, and c are the Weibull parameters; $\Gamma()$ is the gamma function; and $\mathrm{DBH}_{\mathrm{Min}}$, $\overline{\mathrm{DBH}}$, and QMD are the minimum, arithmetic mean, and quadratic mean diameters, respectively. Given that these variables are all well correlated in even-aged stands, equations to predict them from other stand-level variables are easily developed (Fonseca *et al.*, 2009). The utility of this approach is demonstrated several times in long-term growth projections (Cao *et al.*, 1982; Castedo-Dorado *et al.*, 2007b; Newton, 2009). Although the technique is likely more effective in even-aged and single-species stands, Lynch and Moser (1986) demonstrated its usefulness in uneven-aged and mixed-species stands.

When compared to more complex approaches that required a tree list, Qin *et al.* (2007) found that the parameter recovery approach produced similar results in terms of precision, but gave biased predictions of total and merchantable volume. Knowe *et al.* (1997) found that an individual-tree approach gave the best representation of observed diameter distributions because parameter recovery techniques showed a systematic bias with planting density. Similar results were also obtained by Borders and Patterson (1990). However, the technique is effective when a tree list is not available.

An alternative approach is to project a diameter distribution through time by using a theoretical diameter growth equation derived from a diameter probability distribution. This technique was first demonstrated by Bailey (1980), who assumed that trees did not

change their relative sizes over time. The implied diameter growth equation for the Weibull distribution is:

$$\mathrm{DBH}_2 = a_2 + b_2 \cdot \left(\frac{\mathrm{DBH}_1 - a_1}{b_1}\right)^{\left(\frac{c_1}{c_2}\right)} \qquad [10.2]$$

where DBH_i is the DBH and a_i, b_i, and c_i are the Weibull parameters at the first ($i = 1$) and second ($i = 2$) measurements. Cao (1997) further extended this technique to account for mortality. Essentially, the approach involves using parameter recovery to describe the initial stand, projecting changes in stand density with a stand-level equation, determining the diameter distribution parameters after mortality using parameter recovery, and then projecting the growth of each diameter class using the generalized equation of Bailey (1980). The approach was found to generally work better than the parameter recovery technique, but the level of improvements were not statistically significant at the 95% level of confidence (Cao, 1997).

In contrast to relying on a predefined probability density function, Tang *et al.* (1997) used a distribution-independent approach to define relationships between current and future stand diameter distributions. A significant limitation of this approach is that the parameters in equation [10.2] may be a function of age, stand height, or other variables that change with age, which ignoring would result in biased predictions (Oscar Gárcia, personal communication).

10.2.2 Modified stand table projection

Stand table projection is a size-class based approach that involves growing trees by moving them from one diameter class to the next. Various approaches are used to implement stand table projections (see Chapter 4), but all require a method for estimating diameter increment and often mortality for longer projections. However, these equations or the data needed to develop them often are not available. To address this limitation, Clutter and Jones (1980) and Pienaar and Harrison (1988) used stand-level growth and mortality equations to update stand tables. Their approach relied on the assumptions that relative tree size can be used to allocate stand-level mortality, relative tree size generally stays constant over time, and trees within each diameter class are uniformly distributed. However, the approach needed an empirical equation that described changes in relative tree size over time.

Instead of relying on an empirical equation, Nepal and Somers (1992) used the Bailey (1980) method to derive a theoretical diameter increment equation. An algorithm was then used to ensure that basal area and survival matched observed values or predictions made by stand-level equations. Nepal and Somers (1992) found that this technique was superior to Pienaar and Harrison's (1988) method and was still effective in stands with an irregular structure. Cao and Baldwin (1999) further improved the algorithm of Nepal and Somers (1992) by using constrained least squares. However, Trincado *et al.* (2003) found that the

approach of Nepal and Somers (1992) was superior to the one of Cao and Baldwin (1999) when predicting *Eucalyptus* plantations in Chile, particularly for projections longer than four years.

When individual-tree diameter increment and survival equations are available, Cao (2007) showed that a modified, constrained least squares technique works better than the approach of Nepal and Somers (1992). However, this approach would only be applicable if those individual-tree equations were available and a tree list was not. Otherwise, one of the approaches described below would likely be more effective.

10.3 Linked stand- and tree-level models

Individual-tree models are preferred because they maintain information on each tree in a stand, which provides the highest level of resolution and allows various stand structures to be successfully simulated. However, individual-tree models are an example of passive aggregation (Ritchie and Hann, 1997a), which means that changes in stand-level variables like total basal area are estimated from tree-level projections and not simulated directly. This indicates that individual-tree models are unconstrained and may provide unrealistic projections of stand-level attributes. Three primary approaches are used to link stand- and tree-level models: (1) disaggregation; (2) constrained parameters; and (3) combined.

10.3.1 Disaggregation

Disaggregation is the most common means for linking stand- and tree-level models (see review of Ritchie and Hann, 1997a). In this approach, growth and mortality are projected at the stand-level and then allocated to individual trees. The advantages of this approach are that it provides flexibility and numerical compatibility, and stand-level estimates are not influenced by tree-level projections (Ritchie and Hann, 1997a). A variety of approaches (Table 10.1) are used to disaggregate stand growth to individual trees, including: (1) proportional yield (Clutter and Allison, 1974; Somers and Nepal, 1994); (2) proportional increment (Campbell *et al.*, 1979; Moore *et al.*, 1994); (3) constrained least squares (Matney *et al.*, 1990; Cao and Baldwin, 1999); (4) coefficient adjustment (Qin and Cao, 2006); and (5) additive functions (Harrison and Daniels, 1988; Dhôte, 1994).

In proportional yield allocation, tree-level predictions are adjusted proportionally by their predicted final value and the stand-level estimates (e.g. Somers and Nepal, 1994). This concept was widely used in the past for a range of purposes (Clutter and Allison, 1974; Clutter and Jones, 1980; Pienaar and Harrison, 1988; McTague and Stansfield, 1994; 1995). In proportional increment allocation, stand-level increment is disaggregated to individual trees based on tree relative position in the stand and other variables. The technique is used to disaggregate diameter (Leary *et al.*, 1979), basal area (Campbell *et al.*, 1979; Moore *et al.*, 1994), and volume increment (Dahms, 1983; Zhang *et al.*, 1993). For

example, Zhang *et al.* (1993) found that relative size increment was related to a quadratic expression of relative size in the stand:

$$\mathrm{RTG} = b_0 + b_1 \cdot \mathrm{RTS} + b_2 \cdot \mathrm{RTS}^2 \qquad [10.3]$$

where RTG is relative tree increment (tree increment divided by stand total increment), RTS is relative tree size, and the b_is are plot-specific parameters. Zhang *et al.* (1993) indicated that the b_is were dependent on initial number of trees, mean tree size, and the variation of DBH within the stand. However, the limitation of this approach is that it doesn't maintain invariance. Also, Ritchie and Hann (1997b) found that the parameters of equation [10.3] were not significantly related to any stand-level variables, which might explain why it was one of the poorest performing disaggregation approaches they examined.

The constrained least squares method (Matney *et al.*, 1990; Cao and Baldwin, 1999) adjusts tree-level predictions based on constraints of estimated survival, average diameter, and stand basal area from stand-level equations. For example, Qin and Cao (2006) used the following estimator to adjust tree survival probability:

$$\widetilde{\mathrm{PS}}_{i,t+q} = \frac{\left(\widehat{\mathrm{PS}}_{i,t+q} - \left(\sum_j \widehat{\mathrm{PS}}_{j,t+q} - s \cdot \widehat{\mathrm{TPH}}_{t+q}\right)\right)}{n} \qquad [10.4]$$

where $\widetilde{\mathrm{PS}}_{i,t+q}$ is adjusted tree survival probability for the ith tree, t is the beginning of the growth period, q is the length of the growing period, s is the plot size, $\widehat{\mathrm{PS}}_{i,t+q}$ is predicted tree survival probability for the ith tree with a tree-level equation, $\widehat{\mathrm{TPH}}_{i,t+q}$ is predicted number of trees based on a stand-level equation, and n is the number of trees on a plot.

Cao and Baldwin (1999) found that the constrained least squares method significantly improved predictions of future stand development when compared to parameter recovery and stand table projection techniques of Nepal and Somers (1992). The coefficient adjustment method of Qin and Cao (2006) added an additional parameter to each tree-level equation to ensure predictions are identical to those made by stand-level equations.

Additive functions are similar to the coefficient adjustment method of Qin and Cao (2006), but predictions are adjusted rather than model parameters. Examples of additive functions are given in Harrison and Daniels (1988), Dhôte (1994), and Ritchie and Hann (1997b). Additive functions assume that tree increment is linearly related to tree size in any given stand. The relationship has been found to be dynamic and varies from stand to stand, but the changes can be related to stand-level factors. Harrison and Daniels (1988) modeled a stand additivity parameter as a function of site index and dominant height. However, Ritchie and Hann (1997b) found a weak relationship between this parameter and site index and suggested the approach was best suited to young stands (<50 years of age). One limitation of this approach is that it can predict negative increment if not properly specified. The approach of Dhôte (1994) avoided this limitation by employing a truncated distribution of increment such that trees with diameters below an increment threshold are predicted to have increment rates of zero.

Table 10.1 Example of various disaggregation approaches used for predicting tree-level growth, adapted from Ritchie and Hann (1997a).

Source	Dependent variable
Proportional yield	
Clutter and Allison (1974)	Tree basal area
Clutter and Jones (1980)	Tree basal area
Pienaar and Harrison (1988)	Tree basal area
Nepal and Somers (1992)	Tree basal area
Somers and Nepal (1994)	Tree diameter
McTague and Stansfield (1994)	Tree basal area
McTague and Stansfield (1995)	Tree basal area
Qin and Cao (2006)	Tree diameter squared
Proportional increment	
Campbell *et al.* (1979)	Tree basal area increment
Leary *et al.* (1979)	Tree diameter increment

Function[a]

$$\mathrm{ba}_{2i} = \overline{\mathrm{ba}}_2 \cdot \left(\frac{\mathrm{ba}_{1i}}{\overline{\mathrm{ba}}_1}\right)^{\left(\frac{A_2}{A_1}\right)^{0.17192}}$$

$$\mathrm{ba}_{2i} = \mathrm{BA}_2 \cdot \left(\frac{1}{\sum\left(n_{2i} \cdot \frac{\mathrm{ba}_{1i}}{\overline{\mathrm{ba}}_1}\right)^{\left(\frac{A_2}{A_1}\right)^{0.0578109}}}\right) \cdot \left(\frac{\mathrm{ba}_{1i}}{\overline{\mathrm{ba}}_1}\right)^{\left(\frac{A_2}{A_1}\right)^{0.0578109}}$$

$$\mathrm{ba}_{2i} = \mathrm{BA}_2 \cdot \left(\frac{1}{\sum\left(n_{2i} \cdot \frac{\mathrm{ba}_{1i}}{\overline{\mathrm{ba}}_1}\right)^{\left(\frac{A_2}{A_1}\right)^{0.2333}}}\right) \cdot \left(\frac{\mathrm{ba}_{1i}}{\overline{\mathrm{ba}}_1}\right)^{\left(\frac{A_2}{A_1}\right)^{0.2333}}$$

$$\mathrm{ba}_{2i} = g \cdot d_{2i}^2 \cdot \left(\beta_0 \cdot \mathrm{e}^{\left(\beta_1 \cdot \breve{d}_{2i}\right)}\right);\ \breve{d}_{2i} = \mathrm{f}(d_{1i}, \theta)$$

$$d_{2i} = \left(\left(\left(\frac{\delta_f \cdot \left(d_{1f} + \Delta d_f\right) - d_{1f}}{\Delta d_f}\right) \cdot \Delta d_i + d_{1i}\right) \cdot \left(d_{1i} + \Delta d_i\right)^{-1}\right) \cdot \left(d_{1i} + \Delta d_i\right)$$

$$\mathrm{ba}_{2i} = \mathrm{BA}_2 \cdot \frac{\tilde{d}_{2i}}{g \cdot \sum \tilde{d}_{2i}};\ \tilde{d}_{2i} = \mathrm{f}\left(d_{1i}, \frac{\mathrm{BA}_2}{\mathrm{BA}_1}, \frac{N_2}{N_1}, D_{\mathrm{q}1}, t_2 - t_1\right)$$

$$\mathrm{ba}_{2i} = \mathrm{BA}_2 \cdot \frac{\tilde{d}_{2i}^2}{g \cdot \sum \tilde{d}_{2i}^2};\ \tilde{d}_{2i} = \mathrm{f}\left(d_{1i}, \frac{\mathrm{BA}_2}{\mathrm{BA}_1}, N_1, \mathrm{BA}_1, \mathrm{sp}, t_2 - t_1\right)$$

$$d_{2i}^2 = d_{1i}^2 \cdot \left(\frac{40\,000 \cdot s \cdot \mathrm{BA}_2}{\pi \cdot \sum_j^n \left(P_{2j} \cdot d_{2j}^2\right)}\right)$$

$$\Delta \mathrm{ba}_i = \Delta \mathrm{BA} \cdot r \cdot \mathrm{ba}_{1i}^{0.93}$$

$$\Delta d_i = \left(N_2 \cdot \overline{\Delta d}\right) \cdot \left(\kappa_1 + 10^{\beta_0 + \beta_1 \cdot \log\left(\mathrm{GS}_i + \kappa_2\right) + \beta_2 \cdot \log\left(\mathrm{GS}_i + \kappa_2\right)^2 + \beta_3 \cdot \log\left(\mathrm{GS}_i + \kappa_2\right)^3}\right)$$

(*continued*)

Table 10.1 (*Continued*)

Source	Dependent variable
Dahms (1983)	Tree volume increment
Zhang *et al.* (1993)	Tree volume increment
Moore *et al.* (1994)	Tree basal area increment
Qin and Cao (2006)	Tree diameter growth squared
Additive	
Harrison and Daniels (1988)	Tree basal area increment
Dhôte (1994)	Tree basal area increment
Ritchie and Hann (1997b)	Tree basal area increment

[a] Where ba_{2i} is the tree basal area for the ith tree at the end of the growth period, $\overline{ba}_2$ is the mean basal area per tree at the end of the growth period, ba_1 is the tree basal area for the ith tree at the start of the growth period, $\overline{ba}_1$ is the mean basal area per tree at the start of the growth period, A_2 is the stand age at the end of the growth period, A_1 is the stand age at the start of growth period, BA_2 is the stand basal area at the end of the growth period, BA_1 is the stand basal area at the start of the growth period, n_{2i} is the predicted number of trees per unit area represented by each tree at the end of the growth period, g is the conversion of diameter squared to basal area in square units, D_q is the quadratic mean diameter, D_{q1} is the quadratic mean diameter at the start of the growth period, N_2 is the number of trees at the end of growth period, N_1 is the number of trees at the start of the growth period, t_2 is the year at the end of the growth period, t_1 is the year at the start of the growth period, sp is species, P_{2j} is the tree probability of survival, Δba_i is the tree basal area increment, $\overline{\Delta ba}$ is mean tree basal area increment, ΔBA is the stand basal area increment, r is an unspecified proportional adjustment in Campbell *et al.* (1979), β_is are parameters to be estimated, $\overline{\Delta d}$ is the mean diameter increment per tree, Δv_i is the tree volume increment, ΔV is the stand volume increment, d_{1i} is the tree diameter for the ith tree at the start of the growth period,

Function[a]

$$\Delta \mathrm{v}_i = \Delta V \cdot \left(\frac{d_{1i}^{1.818} \cdot h_{1i}^{1.786}}{\sum \left(d_{1i}^{1.818} \cdot h_{1i}^{1.786} \right)} \right)$$

$$\Delta \mathrm{v}_i = \Delta V \cdot \left(\beta_0 + \beta_1 \cdot \frac{\mathrm{v}_{1i}}{V_1} + \beta_2 \left(\frac{\mathrm{v}_{1i}}{V_1} \right)^2 \right)$$

$$\Delta \mathrm{ba}_i = \Delta \mathrm{BA} \cdot \left(\beta_0 + \beta_1 \frac{\mathrm{ba}_{1i}}{\mathrm{BA}_1} + \beta_2 \left(\frac{\mathrm{ba}_{1i}}{\mathrm{BA}_1} \right)^2 \right)$$

$$\tilde{d}_{2i}^2 = d_{1i}^2 + \left(\frac{s \cdot \left(\frac{40\,000 \cdot \mathrm{BA}_2}{\pi} \right) - \sum \left(P_{2j} \cdot d_{1j}^2 \right)}{\sum \left(P_{2j} \cdot \left(d_{2j}^2 - d_{1j}^2 \right) \right)} \right) \cdot \left(\widehat{d}_{2i}^2 - d_{1i}^2 \right)$$

$$\Delta \mathrm{ba}_i = \overline{\Delta \mathrm{ba}} + \left(\beta_1 \cdot H^{\beta_3} \cdot \mathrm{e}^{(\beta_2 \cdot \mathrm{SI})} \right) \cdot \left(\mathrm{ba}_{1i} - \overline{\mathrm{ba}}_1 \right)$$

$$\Delta \mathrm{ba}_i = \overline{\Delta \mathrm{ba}} + \left(\mathrm{f} \left(\sum_{\sigma} d_{1i} \right) \cdot \left(d_{1i} - \overline{d}_{\sigma} \right) \right)$$

$$\Delta \mathrm{ba}_i = \overline{\Delta \mathrm{ba}} + \beta_0 \left(\mathrm{ba}_{1i} - \overline{\mathrm{ba}}_1 \right) + \beta_1 \left(\mathrm{cr}_{1i} - \overline{\mathrm{cr}}_1 \right)$$

d_{2i} is the tree diameter for the ith tree at the end of the growth period, Δd_i is the diameter increment of the ith tree, d_{1j} is the initial diameter of the jth diameter class, d_{2j} is the diameter of the jth diameter class at the end of the growth period, v_{1i} is the tree volume for the ith tree at the start of the growth period, V_1 is the stand volume at the start of the growth period, H is the dominant height, SI is the site index, $\overline{d}_{\sigma}$ is a specified lower diameter limit for growth (all trees below the limit have no growth, and in the summation, σ is the index value for the tree of the specified lower diameter limit for growth), cr_{1i} is the tree crown ratio for the ith tree at the start of the growth period, $\overline{\mathrm{cr}}_1$ is the mean tree crown ratio at the start of the growth period, the κ_is are nonlinear functions of $\overline{\mathrm{ba}}$, θ is an array of parameters, h_{1i} is the total tree height for the ith tree at the start of the growth period, GS_i is $\left(\frac{\overline{d}_{1i}}{\sum_i^{10} \overline{d}_{1i}} \right)$ for diameter classes $i = 1, \ldots, 10$, s is size of the plot (ha), δ_f is a growth adjustment for a single tree, d_{1f} is the initial diameter of some selected tree f, Δd_f is the predicted unadjusted diameter growth of some selected tree, and log is the natural logarithm.

Qin and Cao (2006) compared four different methods of disaggregation (additive functions were not examined) and found that they all produced similar results for a range of projection lengths (4–17 years) in loblolly pine plantations. However, when the disaggregation method of coefficient adjustment was compared to an unconstrained individual-tree model, a significant improvement in predicted stand-level attributes was achieved (Qin and Cao, 2006). In Douglas-fir stands of western Oregon, Ritchie and Hann (1997b) compared six different methods of disaggregation (additive and proportional growth) to an unconstrained individual-tree model. In general, they found that the unconstrained individual-tree approach was superior to any of the tested disaggregation approaches for predicting both tree- and stand-level increment (Ritchie and Hann, 1997b). However, the disparity between the disaggregation and the unconstrained individual-tree approach was less evident after crown ratio was eliminated from the individual-tree equations. In addition, they found an additive allocation approach was far superior to simple proportional allocation for predicting basal area increment (Ritchie and Hann, 1997b).

One limitation of Ritchie and Hann (1997b) is that they only looked at short-term (five-year) projections of increment, and the results may not hold for longer projections because of the predominance of mortality in determining net increment. There is also little evidence of how effective disaggregation approaches are in multi-species and uneven-aged stands. McTague and Stansfield (1995) used the approach in uneven-aged and mixed-species stands, but the projections were not compared to an unconstrained individual-tree approach. Ritchie and Hann (1997b) surmised that disaggregation approaches would work better in even-aged, single-species stands because of the limited capability of stand-level equations in multi-species and uneven-aged stands, but this assertion still remains untested.

10.3.2 Constrained

In contrast to the disaggregation approach of allocating stand-level growth and mortality to individual trees, the constrained approach uses a multi-response parameter estimation technique to optimize tree-level predictions at multiple levels. This approach was introduced by Zhang *et al.* (1997b), who used a pair of seemingly unrelated equations, based on the conceptual relationship between basal area of each individual tree and diameter increments of trees in the same diameter class, to constrain predictions:

$$\Delta \mathrm{DBH}_{ji,t} = \mathrm{f}(x_{ji,t} \cdot \beta) + \varepsilon_{ji,t} \qquad [10.5]$$

$$\mathrm{BA}_{j,t+1} = 0.00007854 \cdot \sum_{i=1}^{n_{j,t+1}} \left(\mathrm{DBH}_{ji,t} + \mathrm{f}(x_{ji,t} \cdot \beta)\right)^2 + \varepsilon_{j,t}$$

where $\Delta \mathrm{DBH}_{ji,t}$ is predicted diameter increment (cm) for the ith tree in the jth diameter class at time t, $x_{ji,t}$ is a vector of tree-level covariates, β is a vector of fixed parameters to be estimated from the data, $\mathrm{BA}_{j,t+1}$ is basal area of the jth diameter class at time $t + 1$, $n_{j,t+1}$ is

the stem density (no. ha^{-1}) and the εs are residual errors assumed to be correlated. Zhang *et al.* (1997b) developed a similar equation for tree survival probability. Results from an unthinned loblolly pine stand indicated that the constrained approach provided predictions of growth nearly identical to an unconstrained model, but improved estimates of total stand basal area after 10 to 12 years of simulations (Zhang *et al.*, 1997b).

Cao (2006) modified the approach of Zhang *et al.* (1997b) by substituting stand-level attributes for diameter-class attributes in both diameter increment and survival equations. This improved predictions of stand-level basal area in loblolly pine plantations compared to the diameter-class constrained approach, but both constrained approaches were found inferior to the disaggregation approach of proportional growth after five years of simulations (Cao, 2006).

10.3.3 Combined

To be effective, disaggregation requires accurate predictions of stand-level growth and mortality, while the constrained approach only influences parameter estimates of the tree-level equations. In both cases, information is not combined from both tree- and stand-levels to improve predictions. The combined approach of Yue *et al.* (2008) uses a composite estimator to join estimates of both tree- and stand-level equations to improve predictions at both levels. The combined approach has three primary steps: (1) preliminary prediction; (2) combination; and (3) feedback modification (Figure 10.3). In the first step, both unconstrained tree- and stand-level growth equations are used to predict basal area growth. These predictions are combined based on a variance and covariance method:

$$\mathrm{BA_C} = \mathrm{BA_S} + \left(\frac{\left(\sigma_\mathrm{S}^2 - \sigma_\mathrm{TS}\right)}{\left(\sigma_\mathrm{T}^2 + \sigma_\mathrm{S}^2 - 2 \cdot \sigma_\mathrm{TS}\right)} \right) \cdot \left(\mathrm{BA_T} - \mathrm{BA_S}\right) \qquad [10.6]$$

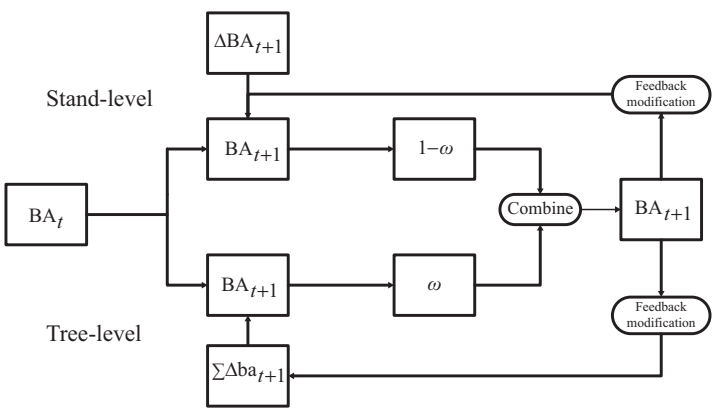


Figure 10.3 Flowchart of the combined stand- and tree-level approach of Yue *et al.* (2008). BA is the stand total basal area, ba is the individual-tree basal area, and ω is a weighting factor.

where BA_C is the combined estimate of total basal area, BA_S is the estimate of total basal area as predicted by the stand-level equation, BA_T is the estimate of total basal area as predicted by the tree-level equation, ${\sigma^2}_S$ is the variance of aggregated variables from the stand-level equation, ${\sigma^2}_T$ is the variance of aggregated variables from the tree-level equation, and σ_{TS} is the covariance of variables in the tree- and stand-level equations. In the final step, the combined estimate of total basal area is disaggregated to individual trees based on the ratio of the combined total basal area and tree-level predicted total basal area:

$$DBH_{i,t+1} = DBH_{i,t} + \Delta DBH_{i,t+1} \cdot \sqrt{\frac{BA_C}{BA_T}} \qquad [10.7]$$

Yue *et al.* (2008) found the gains in efficiency for predicting total basal area using the combined approach compared to the tree- and stand-level equations only were up to 43 and 16%, respectively, on an independent dataset. The combined approach is a relatively new and novel concept for growth modeling that deserves further attention, particularly in modeling individual-tree survival.

10.4 Summary

All models have imposed limitations based on their resolution. Various methods are used to link models of varying resolutions to take advantage of features of each approach. Although the primary methods of combining models are presented in this chapter, a variety of other proposed model combinations are used. For example, Qin *et al.* (2007) found that combining an individual-tree model with a diameter-distribution model was the most effective approach they tested, while Härkönen *et al.* (2010a) found that an unconstrained individual-tree model predicted stand volume better than a constrained model. In addition, as discussed in Chapter 8, several individual-tree growth models like PROGNAUS (Monserud *et al.*, 1997) and ORGANON (Hann, 2011) impose constraints so that the predictions are consistent with a stand-level maximum size–density line.

The combined stand-level and size-class approach of Daniels and Burkhart (1988) is effective when a tree list in not available, but becomes increasingly more difficult as stand structures become more varied due to management or the presence of multiple species. The combined individual-tree and stand-level approach has received significant attention, particularly disaggregation. However, gains using the disaggregation approach are mixed in the limited number of studies that have rigorously examined them, and their performance is likely dependent on an array of factors. Other approaches like the combined approach deserve further attention and should see increased use in the future.